

Exercise (home):

Derive div. theorem starting from Green's theorem:

Tricks:

- write ψ as a function of γ'

$$\rightarrow \text{curl } \phi = \frac{\partial \phi_2}{\partial x} - \frac{\partial \phi_1}{\partial y} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} = \text{div } F$$

7/10/2026

• Corollary 2:

Let $A \subset \mathbb{R}^2$ a regular domain s.t. ∂A is positively oriented.

Let's consider the vector fields $F, G, H: \bar{A} \rightarrow \mathbb{R}^2$ defined as:

$$\bullet F(x, y) = (-y, x), \quad G(x, y) = (0, x), \quad H(x, y) = (-y, 0)$$

$$\text{Then Area}(A) = \frac{1}{2} \int_{\partial A} F \cdot d\mathbf{l} = \int_{\partial A} H \cdot d\mathbf{l} = \int_{\partial A} G \cdot d\mathbf{l}$$

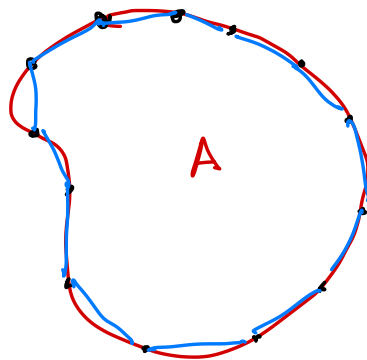
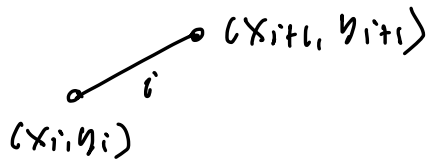
sketch of the proof:

$$\text{curl } F = 2, \quad \text{curl } G = \text{curl } H = 1$$

$$\text{Green's theorem: } \text{Area}(A) = \int_A 1 \, dA = \int_A \underbrace{\text{curl } G}_{=1} \, dA = \int_A \underbrace{\text{curl } H}_{=1} \, dA = \frac{1}{2} \int_A \underbrace{\text{curl } F}_{=2} \, dA$$

$$= \int_{\partial A} G \cdot dl = \int_{\partial A} H \cdot dl = \frac{1}{2} \int_{\partial A} F \cdot dl$$

$$\text{area}(A) = \int_{\partial A} G \cdot dl = \sum_i \int_{S_i} G \cdot dl$$



• Corollary 3:

Let $A \subset \mathbb{R}^2$ be a regular domain whose ∂A is positively oriented. Let $\nu: \partial A \rightarrow \mathbb{R}^2$ the field of outer unit normal vectors on ∂A and let $f: \bar{A} \rightarrow \mathbb{R}$ be a scalar field s.t. $f \in C^2(\bar{A})$. Then:

$$\iint_A \Delta f(x, y) dx dy = \int_{\partial A} \text{grad } f \cdot \nu dl$$

sketch proof:

$$\Delta f(x, y) = \text{div}(\text{grad } f)$$

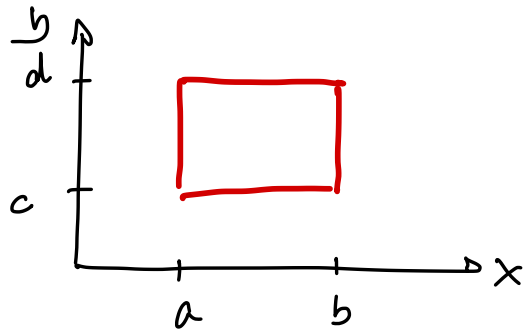
$$\iint_A \text{div}(\text{grad } f) dx dy = \int_{\partial A} (\text{grad } f) \cdot \nu dl$$

Home exercise:

Green's theorem in a restricted setting

We assume that A is $A = (a, b) \times (c, d)$, $a, b, c, d \in \mathbb{R}$

$$a < b, c < d$$



Hint: take a generic field $F \in C^1(\bar{A}, \mathbb{R}^2)$, and try to verify Green's theorem:

$$1) \iint_A \text{curl } F$$

$$2) \iint_{\partial A} F \cdot dl$$

CHAPTER 3: SURFACE INTEGRALS, DIVERGENCE THEOREM, AND STOKES' THEOREM

3.1 Recalls and preliminary notations

3.1.1 change of variables in an integral with several variables

$$\text{let } f: \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}, f \in C^0(\Omega) \\ (x_1, \dots, x_n) \mapsto f(x_1, \dots, x_n)$$

let $\Omega \subset \mathbb{R}^n$ be an open domain

A change of variables is described by a bijective function μ defined as

$$\mu: \hat{\Omega} \subset \mathbb{R}^n \rightarrow \Omega \subset \mathbb{R}^n$$

$$(y_1, \dots, y_n) \mapsto \mu(y_1, \dots, y_n) = (x_1, \dots, x_n)$$

$\hat{\Omega} \subset \mathbb{R}^n$ is an open domain.

we express the old variables $(x_1, \dots, x_n)$ as a function of the new ones $(y_1, \dots, y_n)$.

$$x_1 = \mu_1(y_1, \dots, y_n)$$

$$\vdots$$

$$x_n = \mu_n(y_1, \dots, y_n)$$

μ is defined s.t.

$$\underbrace{\det(\nabla \mu(y))}_{\text{Jacobian}} = \det \begin{pmatrix} \frac{\partial \mu_1}{\partial y_1} & \dots & \frac{\partial \mu_1}{\partial y_n} \\ \vdots & & \\ \frac{\partial \mu_n}{\partial y_1} & \dots & \frac{\partial \mu_n}{\partial y_n} \end{pmatrix} \neq 0 \quad \forall y \in \hat{\Omega}$$

For changing the variables of an integral we do

$$\iint \dots \int_{\Omega} f(x_1, \dots, x_n) dx_1 \dots dx_n = \iint \dots \int_{\hat{\Omega}} f(u(y_1, \dots, y_n)) \|\det \mathcal{M}(y)\| dy_1 \dots dy_n$$

Example 1: cylindrical coordinates

• old variables : (x_1, x_2, x_3) . Cartesian coordinates in $\mathbb{R}^3$

• new variables : $(y_1, y_2, y_3) = (r, \theta, z)$ cylindrical coords in $\mathbb{R}^3$

$$x_1 = u_1(r, \theta, z) = r \cos \theta$$

$$x_2 = u_2(r, \theta, z) = r \sin \theta$$

$$x_3 = u_3(r, \theta, z) = z$$

$$\Omega = \mathbb{R}^3$$

$$\hat{\Omega} \quad , \quad \begin{aligned} r &> 0 \\ \theta &\in [0, 2\pi[\\ z &\in \mathbb{R} \end{aligned}$$

$$\det(\nabla u) = \det \begin{pmatrix} \cos \theta & -r \sin \theta & 0 \\ \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = r$$

$$\|\det(\nabla u)\| = r \quad \text{because } r > 0$$

$$\iiint_{\Omega} f(x_1, x_2, x_3) dx_1 dx_2 dx_3 = \iiint_{\hat{\Omega}} \underbrace{f(u(r, \theta, z))}_{f(r \cos \theta, r \sin \theta, z)} \|\det \nabla u\| dr d\theta dz$$

$$\text{If } f=1, \quad \Omega = \{ (x_1, x_2, x_3) \in \mathbb{R}^3 : x_1^2 + x_2^2 < R^2 \text{ and } x_3 \in (0, h) \}$$

$$\iiint_{\Omega} f dx_1 dx_2 dx_3 = \text{vol}(\Omega) = \pi R^2 h$$

$$= \iiint_{\hat{\Omega}} f(u(r, \theta, z)) \|\det \nabla u\| dr d\theta dz =$$

$$\int_0^R \int_0^{2\pi} \int_0^h 1 \underbrace{r}_{\text{det } \varphi u} d\tau d\theta dr = 2\pi h \int_0^R r dr = \pi R^2 h$$

$$r > 0, \theta \in [0, 2\pi], \varphi \in [0, \pi]$$

Example 2: spherical coords.

$$x_1 = \mu_1(r, \theta, \varphi) = r \sin \varphi \cos \theta$$

$$x_2 = \mu_2(r, \theta, \varphi) = r \sin \varphi \sin \theta$$

$$x_3 = \mu_3(r, \theta, \varphi) = r \cos \varphi$$

$$\det(\varphi u) = -r^2 \sin \varphi$$

$$\|\det \varphi u\| = r^2 \sin \varphi$$

$$\uparrow \begin{matrix} r > 0 \\ \varphi \in [0, \pi] \end{matrix}$$

$$\Omega = \{ (x_1, x_2, x_3) \in \mathbb{R}^3 \text{ s.t. } x_1^2 + x_2^2 + x_3^2 = R^2 \}$$

$$\text{vol}(\Omega) = \frac{4}{3} \pi R^3 = \iiint_{\Omega} 1 dx dy d\tau = \iiint_{\hat{\Omega}} 1 \|\det \varphi u\| = \dots = \frac{4}{3} \pi R^3.$$

2.1.2 Surfaces in $\mathbb{R}^3$

• Notation.

For a field $F: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ we denote

$$F(x, y) = (F_1(x, y), F_2(x, y), F_3(x, y))$$

we denote so $F_{,x}$ $F_{,y}$ $F_{,z}$

$$F_{,x} = \frac{\partial F}{\partial x}, \quad F_{,y} = \frac{\partial F}{\partial y}, \quad F_{,z} = \frac{\partial F}{\partial z}$$

$$F_{\alpha, x} = \frac{\partial F_{\alpha}}{\partial x}, \quad \alpha = 1, 2, 3.$$

• Definition 1:

$\Sigma \subset \mathbb{R}^3$ is called a **regular surface** if:

a) There exists $A \subset \mathbb{R}^2$, a bounded open domain, s.t. ∂A is a (piecewise) closed regular simple curve.

b) There exists a function $\sigma : \bar{A} \rightarrow \mathbb{R}^3$
 $(u, v) \mapsto \sigma(u, v) =$
 $= (\sigma_1(u, v), \sigma_2(u, v), \sigma_3(u, v))$

that satisfies

• $\sigma \in C^1(\bar{A}, \mathbb{R}^3)$, $\sigma(\bar{A}) = \Sigma$ is injective over A .

• the vector $\sigma_{,u} \times \sigma_{,v}$ is s.t.

$$\|\sigma_{,u} \times \sigma_{,v}\| \neq 0 \quad \forall (u, v) \text{ in } A$$

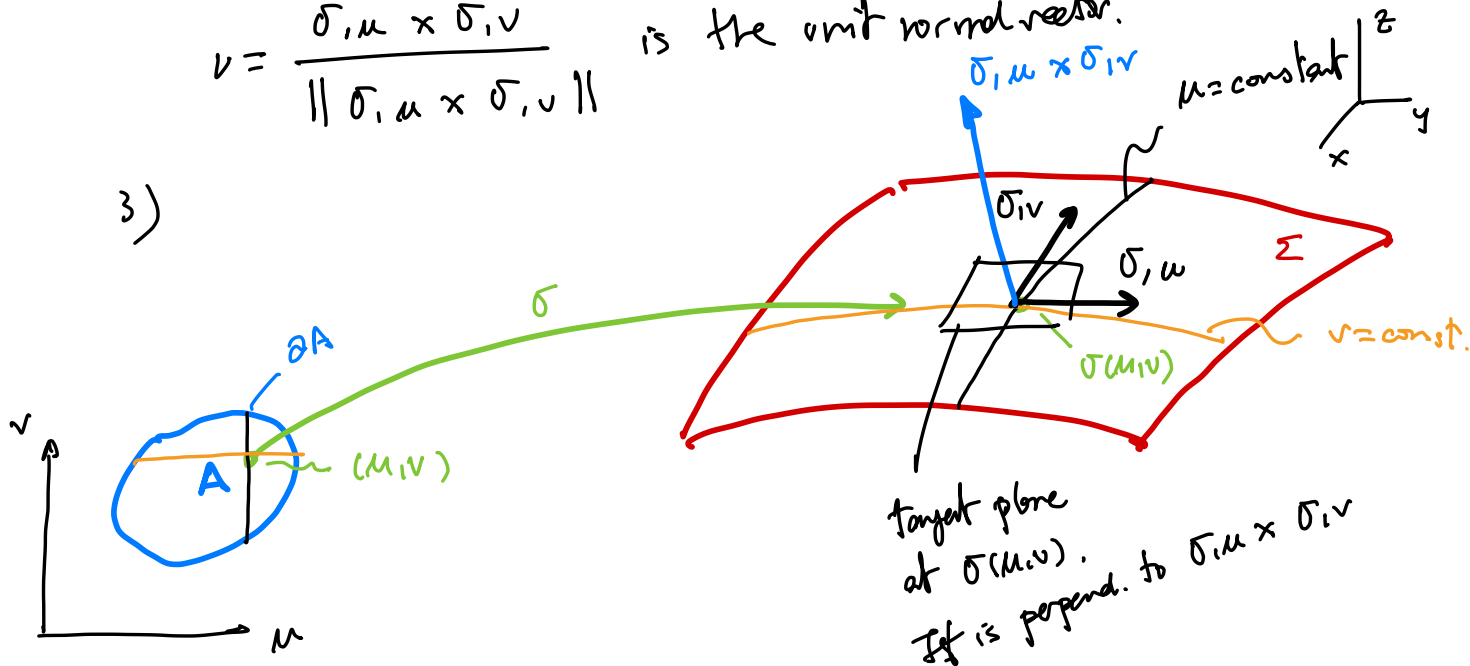
Remarks:

1) σ is a parameterization of Σ .

2) $\sigma_{,u} \times \sigma_{,v}$ is the normal vector of Σ

$$v = \frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|} \quad \text{is the unit normal vector.}$$

3)



4) Analogy with curves in $\mathbb{R}^3$

curve Γ	surface Σ
• $\gamma: [a, b] \subset \mathbb{R} \rightarrow \Gamma \subset \mathbb{R}^3$ $t \mapsto \gamma(t)$ • $\gamma([a, b]) = \Gamma$ injective over $[a, b[$ • $\ \gamma'(t)\ \neq 0 \quad \forall t$ • $dl = \ \gamma'(t)\ dt$	• $\sigma: \bar{A} \subset \mathbb{R}^2 \rightarrow \Sigma \subset \mathbb{R}^3$ $(u, v) \mapsto \sigma(u, v)$ • $\sigma(\bar{A}) = \Sigma$ injective over A • $\ \sigma_u \times \sigma_v\ \neq 0$ • $dA = \ \sigma_u \times \sigma_v\ du dv$

• Definition 2:

we say that $\Sigma \subset \mathbb{R}^3$ is a piecewise regular surface if
 $\exists$ regular surfaces $\Sigma_1, \Sigma_2, \dots, \Sigma_k$, s.t. $\Sigma = \bigcup_{i=1}^k \Sigma_i$.

• Definition 3:

A (piecewise) regular surface is orientable if $\exists$ a field of normals $\nu: \Sigma \rightarrow \mathbb{R}^3$ that is continuous. Such field of normals is called the orientation of Σ .

